

ESWAR KRISHNA P

Mechanical System Integration Engineer (Aspiring)

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PROFILE

Final-year Smart Manufacturing Engineering student (B.Tech, IIITDM Kancheepuram) working toward a career as a Mechanical System Integration Engineer — someone who can take a robotic system from requirement through mechanical design, structural validation, and manufacturing, and into working integration with electronics and software. Over three years, has independently driven 15+ projects spanning mechanical design (Fusion 360), structural analysis and FEA (ANSYS Workbench), and embedded/robotics integration (ROS2, Gazebo, URDF, Arduino, Raspberry Pi). Mechanical Lead of a college robotics club; currently completing a robotics research internship rehabilitating a legacy autonomous platform. Seeking a 6-month internship in mechanical design, structural analysis, or robotics system integration.

EDUCATION

B.Tech, Smart Manufacturing Engineering — IIITDM Kancheepuram

2023 – 2027 (Expected)

CGPA: 7.03 | Relevant Coursework: Robotics & Automation, Multibody System Dynamics, Mechatronic Systems, Sensors & Controls, Theory of Machines, Electrical Drives, Intelligent Robot Systems, Finite Element Analysis, Additive Manufacturing

Intermediate (12th) — Sasi Junior College — 97.8% | 10th — 99.8%

2021 – 2023

ENGINEERING EXPERIENCE

Design Intern | STAM Digital Tooling

2025 (1 Month)

- Designed 3D and 2D CAD models for injection mould components used directly in production, working daily in Fusion 360 on core, cavity, inserts, plates, and standard mould components.
- Prepared manufacturing/engineering drawings and coordinated with production engineers to understand machining requirements, gaining exposure to CNC turning, milling, and mould assembly.
- Studied complete mould architecture — gating systems, ejection mechanisms, cooling channels, parting lines, and tolerances — and learned to evaluate designs against machinability, assembly access, and tolerance, not just geometry.

Robotics Research Intern — Pioneer 3-AT Rehabilitation | Robotics and Automation Laboratory (Guide: Prof. Sreekumar)

2026

(Ongoing, 3 Months)

- Rehabilitating a legacy Pioneer 3-AT autonomous mobile robot, unused for years, back to an operational state for research applications — covering mechanical condition assessment, battery replacement planning, and controller/hardware verification.
- Initial instinct was to replace the electronics entirely; under guidance, shifted to first fully understanding the existing SH2 controller architecture and documenting it, given limited legacy documentation — a systems-engineering lesson in preserving before replacing.
- Learning and applying ROS2, Ubuntu/Linux, and building URDF models for Gazebo simulation; planning future LiDAR integration and autonomous campus navigation as research outcomes.

FLAGSHIP ROBOTICS & MECHANICAL DESIGN PROJECTS

Agribot Mobile Base — Technoxian (Mechanical Lead: Drivetrain & Chassis)

(Ongoing)2026

- Owned the complete mechanical design of a skid-steer mobile base for an agricultural robot required to move over uneven terrain within strict weight and power limits.
- Ran the full design cycle — requirement analysis, concept generation, motor/torque calculations, wheel and shaft/bearing selection, timing-belt transmission design, CAD modelling, assembly, procurement, manufacturing planning, testing, and iteration.
- Compared direct-drive against HTD timing-belt reduction by calculating how wheel diameter changes affected torque and by distinguishing continuous vs. peak motor torque; selected belt-reduction after weighing power constraints and safety factor.
- Modelled the chassis in aluminum extrusion (Fusion 360) and coordinated with electronics and manipulator teams for full-system integration.

Mobile Manipulator (MoMA) — Industrial Wall-Finishing Application & Stability Analysis

2025 – 2026

- Independently developed a conceptual mobile manipulator for an industrial wall-finishing company assignment, aiming to reduce manual finishing effort through an autonomous platform — covering complete mechanical design, CAD modelling, robot configuration, and component selection.
- Performed reachability and workspace study, plus centre-of-gravity stability analysis, calculating safe acceleration limits to prevent overturn under worst-case load configurations across mobile base, telescopic mast, and arm mounting.

- Delivered a conceptual design satisfying company requirements, reinforcing system-level thinking across mechanical, electronic, and software layers.

RoboWars — 8kg & 15kg Combat Robot (Mechanical Design & Weapon Validation) 2025

- Designed chassis and drivetrain in Fusion 360 for competitive combat robots across both weight categories; robot reached complete design and manufacturing and competed in multiple competitions.
- Qualified for and competed at IIT Madras RoboWars, 15 kg category.

2-DOF Robotic Arm — Multibody Dynamics & Kinematics 2025

- Derived forward and inverse kinematics for a 2-degree-of-freedom robotic arm through mathematical modelling, implemented in MATLAB, and verified motion in RoboAnalyzer.
- Carried the design through to a physical prototype with servo-motor integration and mechanical testing — connecting theoretical kinematics to a working system.

STRUCTURAL ANALYSIS & FEA PROJECTS

Cantilever Load Cell Beam — Finite Element Analysis 3rd Year (FEA Course Project)

- Designed and modelled a load-cell beam (Fusion 360) used in industrial weighing systems, then analyzed it in ANSYS Workbench under a fixed-support/400 N end-load boundary condition.
- Generated total deformation, equivalent stress, and equivalent strain results; cross-validated against analytical hand calculations, achieving close agreement and confirming stress concentration at the neck region.
- Refined mesh quality and iterated boundary conditions multiple times to reach accurate, trustworthy results — reinforcing that FEA output is only as good as its setup.

Acrylic Coating Impact-Protection Study — RoboWars 15kg Chassis (Comparative FEA) 2025

- Investigated whether a thin acrylic coating on the outer walls of a 15 kg combat robot could reduce impact stresses and improve structural protection during collisions.
- Built the structural model in ANSYS Workbench with structural steel as the base chassis material, then applied a thin acrylic coating to the external surface; defined impact loading and appropriate boundary conditions for both cases.
- Ran comparative impact simulations for the chassis with and without the acrylic coating, comparing Equivalent (Von Mises) Stress, total deformation, and overall structural response between the two.
- Found acrylic coating effective as a lightweight protective layer for absorbing impact energy and reducing stresses transmitted to the chassis — insight into improving durability without a significant weight increase.

Gujarat Robofest — Mobile Base Chassis Design & Structural FEA 2025

- Designed a complete lightweight mobile-robot chassis in Fusion 360 for competition qualification — chassis layout, component placement, motor positioning, wheel arrangement, and structural frame — prioritizing a low centre of gravity, modular construction, and easy assembly.
- Independently performed structural validation of the chassis under multiple loading conditions (material assignment, meshing, constraints, force application) in ANSYS Workbench and confirmed the design was structurally acceptable.

EMBEDDED & ELECTRONICS PROJECTS

Smart LPG Gas Cylinder Stand — Leak Detection & Weight Monitoring 2024

- Built a real, working interdisciplinary hardware prototype for domestic gas-leakage detection, cylinder-weight monitoring, and automatic alerting, integrating an MQ2 gas sensor, a load cell with HX711 amplifier on Arduino/ESP32.
- Implemented sensor calibration and control logic; the prototype successfully detected gas leaks, read cylinder weight, triggered buzzer/SMS alerts, and activated an exhaust fan.

ShockGuard — Predictive Electrical Safety Node 2024 – 2025

- Designed an electrical-safety prototype aimed at isolating supply before a shock occurs, rather than reacting after an RCCB trips — using capacitive sensing, Arduino-based control logic, and relay-based isolation.
- Handled circuit testing, sensor calibration, and hardware debugging end-to-end, strengthening electronics-side understanding alongside the mechanical core skill set.

FOUNDATIONAL PROJECTS

Line Follower Robot Robotics Club — Early Learning Project

- Independently designed the complete mechanical layout of an IR-sensor-based autonomous line-following robot — chassis in CAD, component/battery placement, sensor positioning, geared DC motor selection, wheel alignment, ESP-based controller integration, assembly, and testing.

Hostel Building CAD Model

Additive Manufacturing Course, 2025

- Converted an existing hostel building's 2D floor plan into a fully scaled 3D CAD model in Fusion 360, using canvas calibration to match real-world measurements, then prepared and validated the model for 3D printing with surface finishing.

8 kg Combat Robot

Team Project

- Handled the mechanical subsystem — chassis understanding, assembly, fabrication coordination, component integration, and weight management — through to complete design and manufacturing; competition participation was later cancelled due to budget limitations.
- Key engineering lesson carried forward: completing a design is one milestone, but field deployment is an entirely separate challenge.

LEADERSHIP

Mechanical Lead | Robotics Club, IIITDM Kancheepuram

2025 – 2026

- Own complete mechanical subsystem planning for competitive robotics platforms — chassis concept finalization, manufacturing feasibility, weight reduction, motor/wheel selection, shaft calculations, timing-belt transmission, bearing arrangement, and component mounting.
- Run design reviews, coordinate procurement planning and timelines, and maintain design documentation across projects.
- Mentor junior members in CAD modelling, mechanical fundamentals, and FEA basics; conduct robotics workshops for newer students, and coordinate directly with electronics and software teams to keep designs integration-ready.

TECHNICAL SKILLS

Mechanical Design & System Integration: Drivetrain layout, skid-steering systems, motor/torque calculations, belt transmission design, forward & inverse kinematics, mounting and packaging design

Structural Analysis & Simulation: ANSYS Workbench (static structural, meshing, boundary conditions, safety-factor validation), RoboAnalyzer, ADAMS, Simulink, MATLAB

CAD & Manufacturing: Fusion 360, injection mould CAD, CNC machining (turning & milling), 3D printing (Cura)

Robotics & Embedded Integration: ROS2, Gazebo, RViz, URDF, Ubuntu/Linux, Arduino, ESP32, Raspberry Pi, MQ2/load-cell/HX711 sensor integration, GSM (SIM800) and RF communication

Programming : Python, C, SQL

Currently Strengthening: Serial and robot communication protocols, control systems, LiDAR, autonomous navigation, industrial robotics, advanced/dynamic FEA, vibration analysis, design validation

CAREER FOCUS

Targeting Mechanical System Integration Engineer roles across mobile robotics, autonomous ground vehicles, warehouse and industrial robots, service robots, and agricultural robots — specifically drivetrain systems, chassis development, manipulator integration, mechanical validation, and full system integration.